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Routing in IOS

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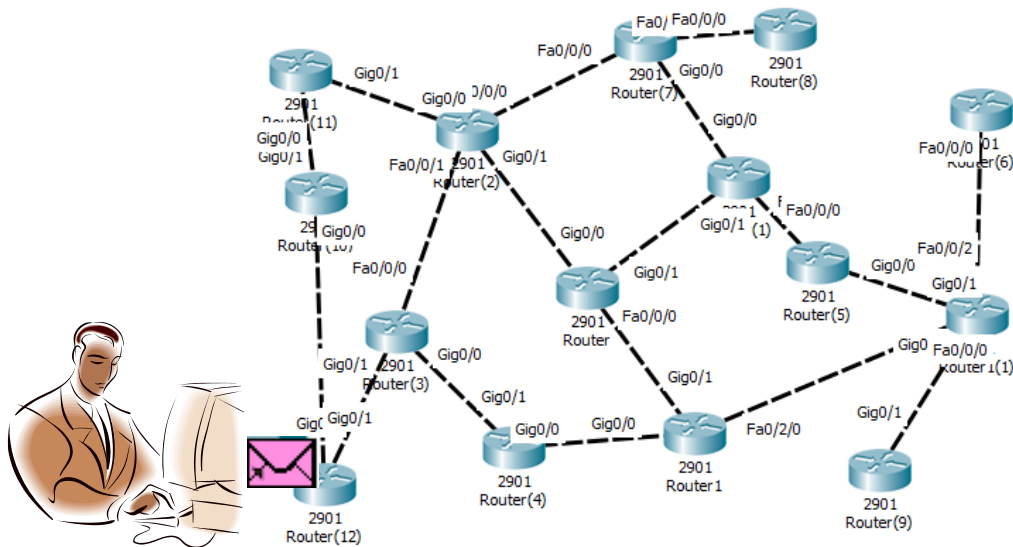
Routing - definition

Routing (trasowanie)

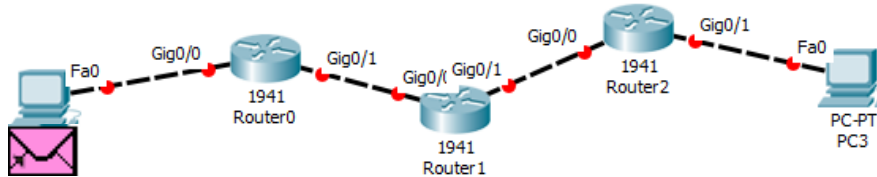
Routing is...

... finding the best routes on the web.

Router task



Network communication



APPLICATION				APPLICATION
PRESENTATION				PRESENTATION
SESSION				SESSION
TRANSPORT				TRANSPORT
	R1	R2	R3	
NETWORK	NETWORK	NETWORK	NETWORK	NETWORK
DATA LINK	DATA LINK	DATA LINK	DATA LINK	DATA LINK
PHYSICAL	PHYSICAL	PHYSICAL	PHYSICAL	PHYSICAL

IPv4 address, netmask^{net}

IP address 192.168.1.45			
192	168	1	45
11000000	10101000	00000001	0101101

Net mask 255.255.255.0 lub <u>_._._._._/24</u>			
255	255	255	0
11111111	11111111	11111111	00000000



Class addressing

Class	First octet	IP address range	Net mask	Number of networks	Number of hosts
A	0 _ _ _	<u>0</u> .0.0.0 - <u>127</u> .255.255.255	<u>255</u> .0.0.0	127	16.777.214
B	1 0 _ _	<u>128</u> .0.0.0 - <u>191</u> .255.255.255	255.255.0.0	16.382	65.534
C	1 1 0 _	<u>192</u> .0.0.0 - <u>223</u> .255.255.255	<u>255.255.255</u> .0	2.097.150	254
D	1 1 1 0	<u>224</u> .0.0.0 - <u>239</u> .255.255.255	for broadcast transmission		
E	1 1 1 1	<u>240</u> .0.0.0 - <u>255</u> .255.255.255	Reserved for IETF		

VLSM - Subnet range for the class B

Net mask: 255.255.0.0 /16

Subnet mask			Number of subnet addresses	Number of host addresses
255.255.192.0	/18	.11000000.00000000	2	16.382
255.255.224.0	/19	.11100000.00000000	6	8.190
255.255.240.0	/20	.11110000.00000000	14	4.094
255.255.248.0	/21	.11111000.00000000	30	2.046
255.255.252.0	/22	.11111100.00000000	62	1.022
255.255.254.0	/23	.11111110.00000000	126	510
255.255.255.0	/24	.11111111.00000000	254	254
255.255.255.128	/25	.11111111.10000000	510	126
255.255.255.192	/26	.11111111.11000000	1.022	62
255.255.255.224	/27	.11111111.11100000	2.046	30
255.255.255.240	/28	.11111111.11110000	4.094	14
255.255.255.248	/29	.11111111.11111000	8.190	6
255.255.255.252	/30	.11111111.11111100	16.382	2

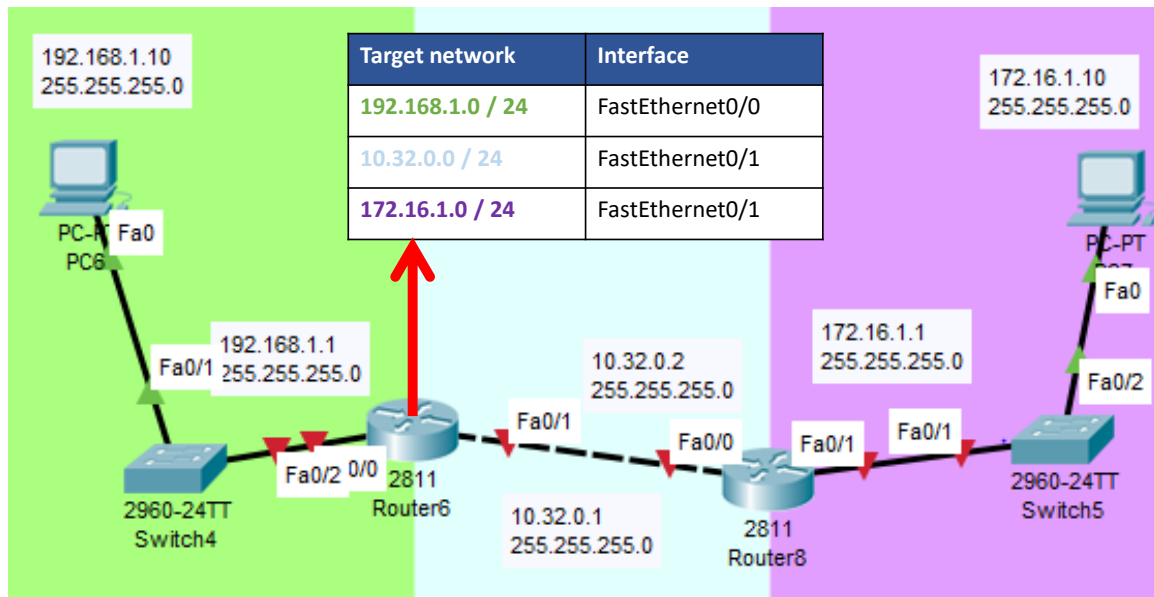
Local addresses

Class	IP address range	Net mask	CIDR notation
A	10.0.0.0 – 10.255.255.255	255.0.0.0	10.0.0.0/8
B	172.16.0.0 – 172.31.255.255	255.240.0.0	172.16.0.0/12
C	192.168.0.0 – 192.168.255.255	255.255.0.0	192.168.0.0/16

Kullanabildiklerimiz

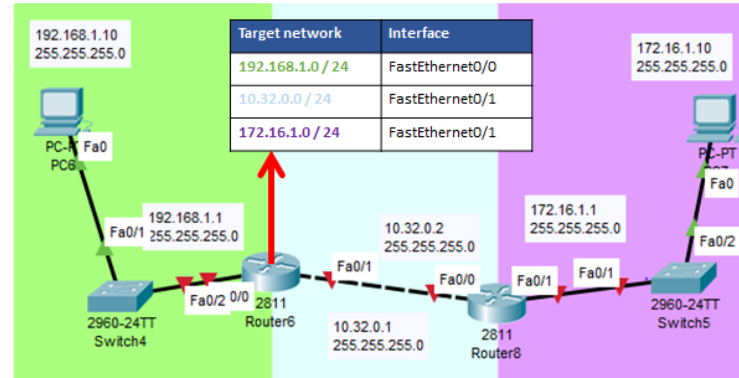
Routing table overview

Routing table



Routing table

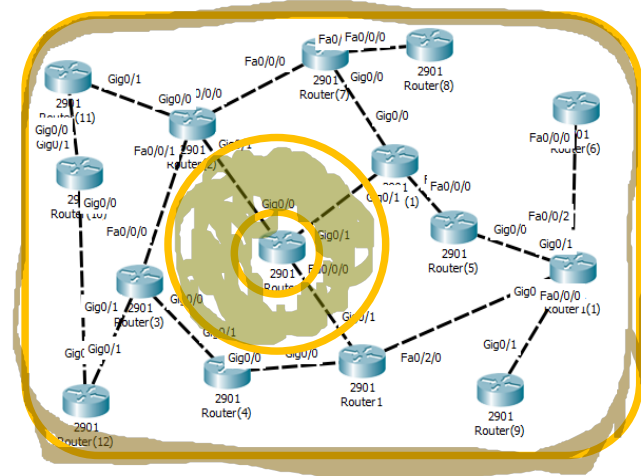
- Table in the memory of a network device (e.g. router),
- Stores information about routes for specific purposes on the web,
- It helps the router determine where to send packets



Routing table rules

The routing table contains:

1. Directly connected networks,
2. Remote networks - static and dynamic,



Routers:

- Each router works independently on the basis of the routing table,
- When one router has network information, it does not mean that the others have it,
- One-way route information does not indicate the return route,

Types of routing

1. Static Routing

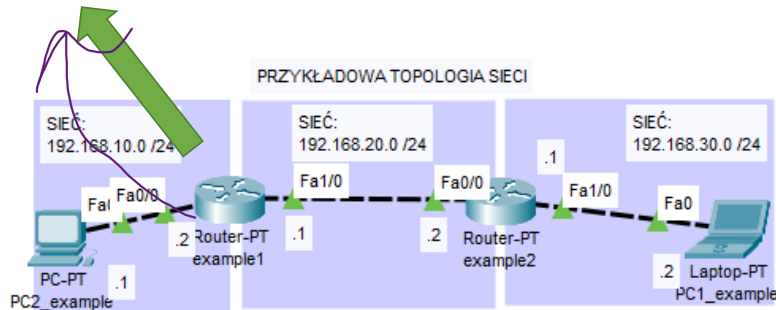
- Manual route entries by the system administrator
- Knowledge of network topology required

2. Dynamic routing

- Routes determined automatically as a result of routing protocols
- No knowledge of network topology necessary

Sample topology - address configuration

```
Router> enable
Router# configure terminal
Router(config)# hostname RXY
RXY(config)# interface fa0/0 ! shorter
RXY(config-if)# ip address 192.168.20.2 255.255.255.0
RXY(config-if)# no shutdown
RXY(config-if)# end
RXY# show running-config
RXY# show ip interface fa0/0
RXY# disable
RXY> exit
```



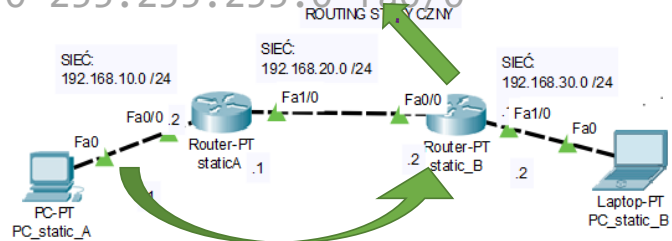
Static Routing

Static route...

...it is a manual entry in the routing table
(prefix and next jump)

Static Routing - Useful Commands

```
R_B> enable
R_B# show interfaces
R_B# show ip interface brief
R_B# show ip route
R_B# configure terminal
R_B(config)# ip route 192.168.10.0 255.255.255.0 192.168.20.1
R_B(config)# ! ip route 192.168.10.0 255.255.255.0 fa0/0
R_B(config)# end
R_B# show ip interface brief
R_B# show ip route
R_B# disable
```



Display the routing table in IOS

R_B> show ip route

```
Router>
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.10.0/24 [1/0] via 192.168.20.1
C    192.168.20.0/24 is directly connected, FastEthernet0/0
C    192.168.30.0/24 is directly connected, FastEthernet1/0

Router>
```

Static Routing - route display

S 192.168.10.0 [1/0] via 192.168.20.1

1

2

3

4

1. **S** - routing table code for a static route
2. **192.168.10.0** - destination route network address
3. **[1/0]** - administrative distance / metric
4. **via 192.168.20.1** - next-hop router IP address,
in this case FastEthernet 1/0 interface IP address of the second router

129 → dynamic

1. Routing table codes

- **C** (connected) - networks connected directly,
- **S** (static) - static networks,
- **I** (IGRP) - IGRP dynamic routing
- **R** (RIP) - RIP dynamic routing
- ...

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, O - OSPF other
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ISU - IS-IS
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.10.0/24 [1/0] via 192.168.20.1
C    192.168.20.0/24 is directly connected, FastEthernet0/0
C    192.168.30.0/24 is directly connected, FastEthernet1/0
```

3. Administrative distance

- The administrative distance (AD) determines the routing source precedence.

An integer between 0 and 255

0 - Directly connected network

1 - Static route

5-100 - for various routing protocols

255 - distance unknown

```
Router>show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP,
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area,
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2,
       E1 - OSPF external type 1, E2 - OSPF external type 2,
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area,
       * - candidate default, U - per-user static route, o - OGP,
       P - periodic downloaded static route
```

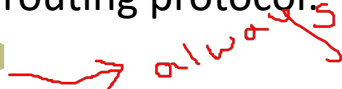
```
Gateway of last resort is not set
```

```
S    192.168.10.0/24 [1/0] via 192.168.20.1
C    192.168.20.0/24 is directly connected, FastEthernet0/0
C    192.168.30.0/24 is directly connected, FastEthernet1/0
```

3. Metrics

A **metric** is a quantitative value that indicates the distance to a given network. The route with the lowest metric is preferred.

Metrics for each routing protocol:

- **0 - static route,**  always
- RIP - number of jumps,
- IGRP and EIGRP - bandwidth, delay, reliability and load,
- IS-IS and OSPF - cost,

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, O - OSPF
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.10.0/24 [1/0] via 192.168.20.1
C    192.168.20.0/24 is directly connected, FastEthernet0/0
C    192.168.30.0/24 is directly connected, FastEthernet1/0
```

4. Next hop

Possible options:

- Next-hop router IP address,
- Output interface leading to the network (FastEthernet)

```
Router>show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 3 subnets
S      172.16.10.0 [1/0] via 172.16.20.1
C      172.16.20.0 is directly connected, FastEthernet0/0
C      172.16.30.0 is directly connected, FastEthernet1/0
```

Summarization on the example of static routing

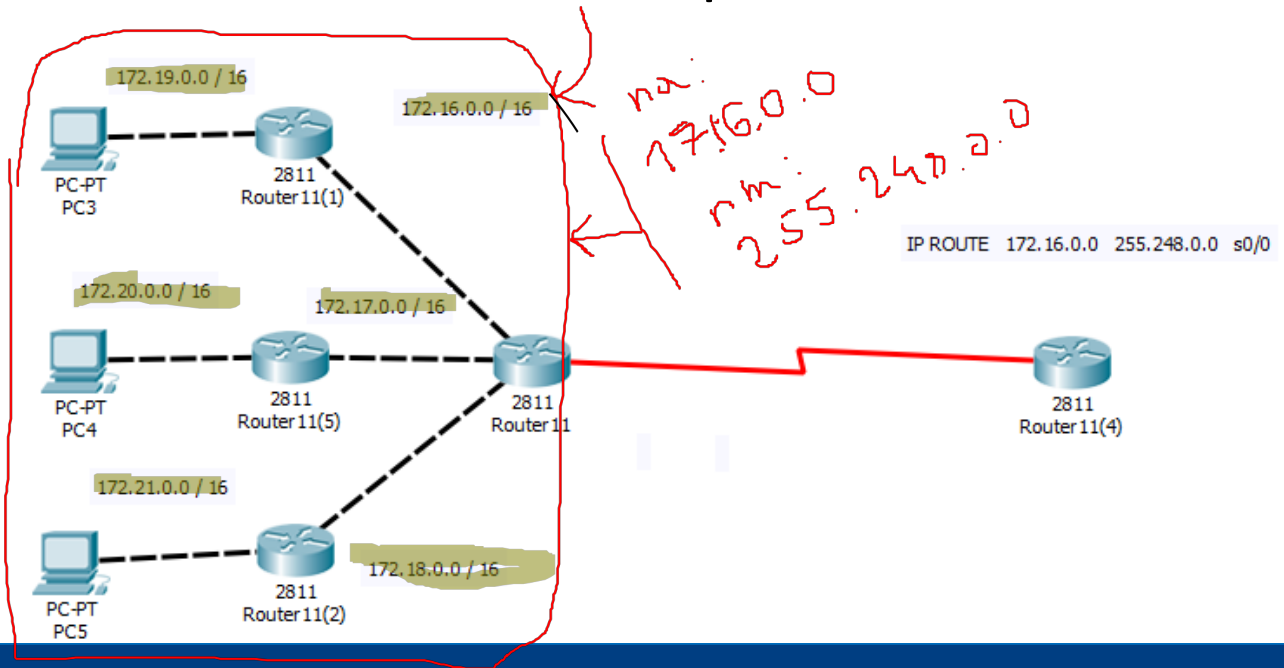
Network summarization

Network summarization (network aggregation) is a single route used to represent multiple routes.

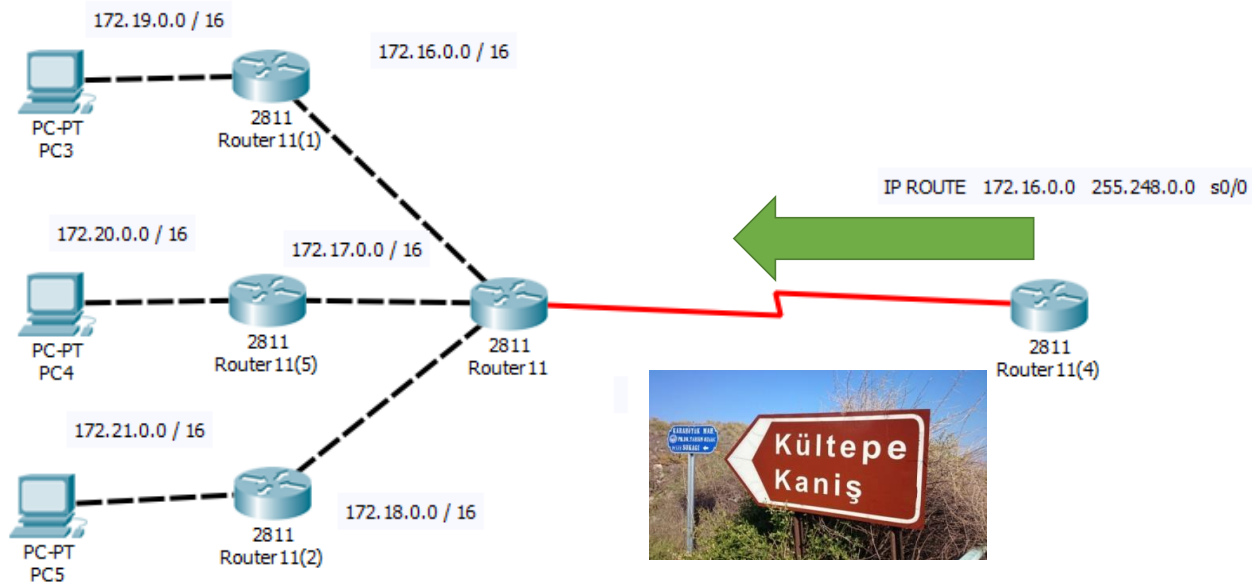
Routes are summarized when:

- target networks can be represented in the form of one summary network,
- destination networks are accessed via the same exit interface or next-hop address.

Network summarization – problem definition



Network summarization



Network summarization

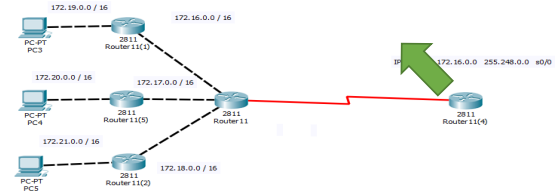
IP address	IP address - binary	
172.16.0.0 /16	10101100	00010000
172.17.0.0 /16		00010001
172.18.0.0 /16		00010010
172.19.0.0 /16		00010011
172.20.0.0 /16		00010100
172.21.0.0 /16		00010101
	172	0

Summarized network:

172.16.0.0 /13

Net mask:

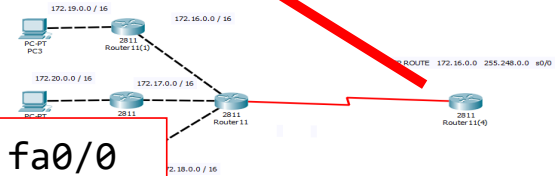
255.248.0.0



Summary static routes

```
RXY> ip route 172.16.0.0 255.255.0.0 fa0/0  
RXY> ip route 172.17.0.0 255.255.0.0 fa0/0  
RXY> ip route 172.18.0.0 255.255.0.0 fa0/0  
RXY> ip route 172.19.0.0 255.255.0.0 fa0/0  
RXY> ip route 172.20.0.0 255.255.0.0 fa0/0  
RXY> ip route 172.21.0.0 255.255.0.0 fa0/0
```

```
RXY> ip route 172.16.0.0 255.248.0.0 fa0/0
```



Other uses of static routes

Static routing is also used to:

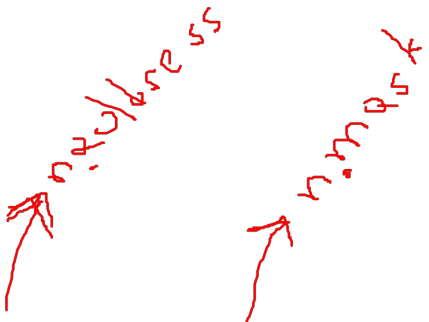
- Define a floating route - backup address for dynamic routing,
- Drop packets - when packets reach the wrong network,

Default static route

The default static route is one where all packets match

Zero route:

- Network address 0.0.0.0
- Netmask 0.0.0.0



```
RXY(config)# ip route 0.0.0.0 0.0.0.0 172.16.20.1
RXY(config)# ip route 0.0.0.0 0.0.0.0 fastEthernet 0/0
```

Default static route

To avoid overlapping the default static routes, the metric of one route is lowered in relation to another.

Default static route metric (0)

```
RXY(config)# ip route 0.0.0.0 0.0.0.0 192.18.55.1
```

Modified administrative distance of static route (254)

```
RXY(config)# ip route 0.0.0.0 0.0.0.0 172.16.20.1 254
```

Default static route

```
Router(config)#
Router(config)#do sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

```
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
```

```
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.20.20.0/30 is directly connected, Tunnel0
L    20.20.20.1/32 is directly connected, Tunnel0
128.0.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    128.0.0.0/16 is directly connected, Serial0/2/0
L    128.0.0.1/32 is directly connected, Serial0/2/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/1.1
L    192.168.1.1/32 is directly connected, GigabitEthernet0/1.1
S    192.168.5.0/24 [1/0] via 20.20.20.2
S    192.168.6.0/24 [1/0] via 20.20.20.2
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/1
L    192.168.10.1/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 is directly connected, Serial0/2/0
```

SVI – EXTRA TASK

Switch – configuration of virtual IP (SVI)

```
Switch> enable
Switch# configure terminal
Switch(config)#
Switch(config)# interface vlan1
Switch(config-if)# ip address 192.168.0.1 255.255.255.0
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)# ip default-gateway 192.168.0.254
Switch(config)# end
Switch# do sh ip int br
Switch# disable
Switch>
```

przełącznik



2960-24TT
Switch0